

```

root_path = fileparts(mfilename("fullpath"));
data_path = fullfile(root_path, 'sym4D_cutsAndFits');

if ~exist('cuts2fit200', 'var')
    ld = load(fullfile(data_path, 'multicuts_fit_dataGH_ei200meV.mat'));
    cuts2fit200 = ld.cuts2fit200;
end
used_ds = containers.Map()

```

```
used_ds =
```

```
Map with properties:
```

```

    Count: 0
   KeyType: char
  ValueType: any

```

```

sel_info = struct('cuts',[],'en_range',[],'fname','','bg_par',[]);
%
sel_info.cuts = cuts2fit200.cuts_dir001FFfit(1);
sel_info.en_range = 30:10:110;
sel_info.fname='EnFit_Ei200ref001off110_2D_constFFCut';
sel_info.bg_par= {};
used_ds('Ei200Br110dir001')=sel_info;
%
sel_info.cuts = cuts2fit200.cuts_dir001FFfit(2);
sel_info.en_range = 30:10:130;
sel_info.fname='EnFit_Ei200ref001off200_2D_constFFCut';
sel_info.bg_par= {};
used_ds('Ei200Br200dir001')=sel_info;
%
sel_info.cuts = {cuts2fit200.cut200_sel};
sel_info.en_range = 40:10:140;
sel_info.fname='EnFit_Ei200Br200dir010_2peaks_2D';
sel_info.bg_par= {};
used_ds('Ei200Br200dir010_2peaks')=sel_info;
%
sel_info.cuts = {cut(cuts2fit200.cut200_sel,[0,0.02,1],[])};

```

```

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 4690699 pixels -- processing data... -----> retained 4604422 pixels
-----
Number of points in input object: 8259853
Fraction of object selected: 56.79% (= 4690699 points) ; Access speed: 0.96GB/sec
Fraction of object retained: 55.74% (= 4604422 points)

**** Total time in cut(sqw) : 0.8sec Including:
Data access time fraction: 39.5%; Transf. time frac.: 38.1%; Resulting pix preparation: 22.3%
-----

```

```

sel_info.en_range = 40:10:140;
sel_info.fname='EnFit_Ei200Br200dir010_1peak_2D';
sel_info.bg_par= {};
used_ds('Ei200Br200dir010_1peak')=sel_info;
%
sel_info.cuts = {cuts2fit200.cut110_sel};
sel_info.en_range = 30:10:110;

```

```

sel_info.fname='EnFit_Ei200Br110dir010_2D';
sel_info.bg_par= {};
used_ds('Ei200Br110dir010')=sel_info;

```

```

%cuts_list = cuts2fit200.cutsS_list();
%cuts_list = {cuts2fit200.cut200_200sel};
%cuts_list = {cuts2fit200.cut110_sel};

```

```

select= 'Ei200Br200dir010_2peaks';
sel_info = used_ds(select);

```

```

cuts_list = sel_info.cuts

```

```

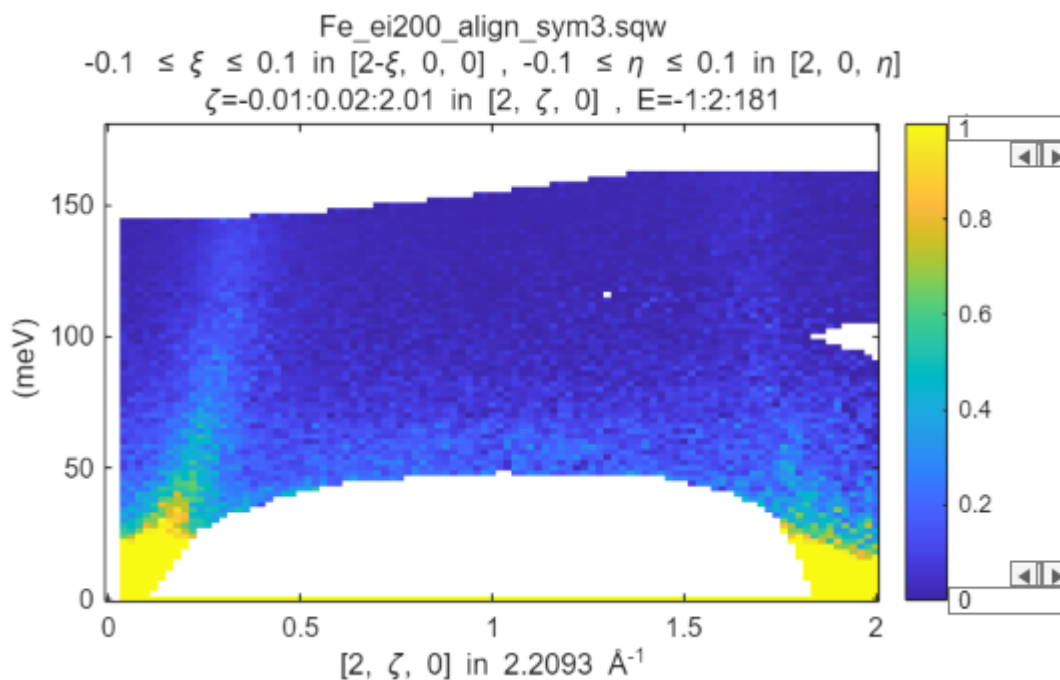
cuts_list = 1×1 cell array
    {1×1 sqw}

```

```

mi = maps_instrument(200,600,'S');
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
for i=1:numel(cuts_list)
    cuts_list{i} = cuts_list{i}.set_instrument(mi);
    cuts_list{i} = cuts_list{i}.set_sample(sample);
    plot(cuts_list{i}); lz 0 1; keep_figure;
end

```



```

dir_name = "GH";

```

```

dE_step = 2; %original energy transfer step data were binned to. No point in
going finer
half_dE = 10; % half width of data binning
%cut_en = 30:10:120;
%cut_en = 40:10:150;
%cut_en = 145;

```

```
cut_en = sel_info.en_range;

bg_par = sel_info.bg_par;
cuts_list{end+1} = bg_par;
fbase_name = sel_info.fname;
```

```
[fit_res_200fix_bg,res_name] = fit_multicuts_along_direction(...
    cuts_list,fbase_name,dir_name,cut_en,dE_step,half_dE,true);
```

```
*****
```

```
**** En = 40±10
```

```
*****
```

```
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
```

```
*** Step 1 of 1; Read data for 267498 pixels -- processing data... -----> included 239177 pixels
```

```
-----
Number of points in input object: 267498
```

```
Fraction of object selected: 100.00% (= 267498 points) ; Access speed: 1.85GB/sec
```

```
Fraction of object retained: 89.41% (= 239177 points)
```

```
**** Total time in cut(sqw) : 0.0sec Including:
```

```
Data access time fraction: 38.9%; Transf. time frac.: 59.0%; Resulting pix preparation: 0.0%
```

```
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
```

```
*** Step 1 of 1; Read data for 267498 pixels -- processing data... -----> retained 239177 pixels
```

```
-----
Number of points in input object: 267498
```

```
Fraction of object selected: 100.00% (= 267498 points) ; Access speed: 1.60GB/sec
```

```
Fraction of object retained: 89.41% (= 239177 points)
```

```
**** Total time in cut(sqw) : 0.1sec Including:
```

```
Data access time fraction: 21.2%; Transf. time frac.: 53.9%; Resulting pix preparation: 24.8%
```

```
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
```

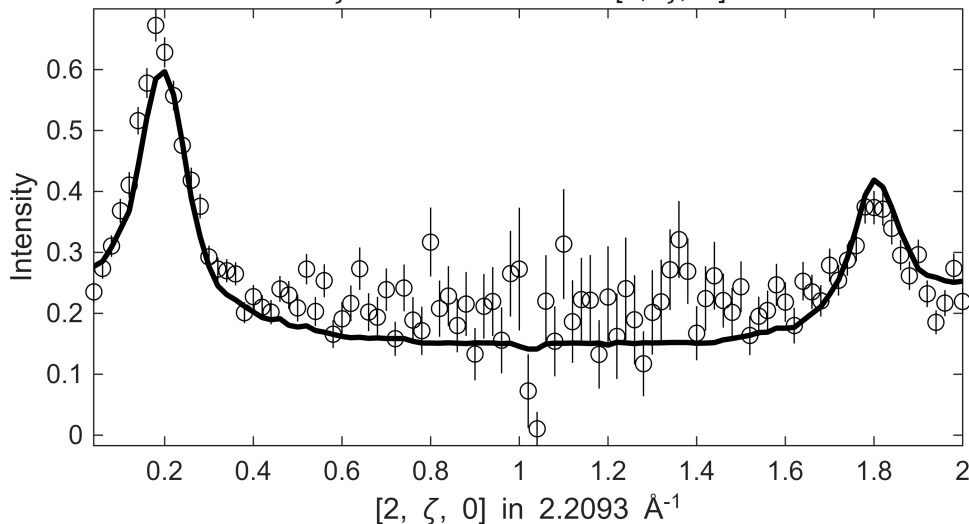
```
*** Step 1
```

Fe_ei200_align_sym3.sqw

S=0.63±0.044; J0= 23±0.37; gamma=7.03964±1.28302

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 30 ≤ E ≤ 50

ζ=-0.01:0.02:2.01 in [2, ζ, 0]



```
of 1; Read data for 267498 pixels -- processing data... -----> retained 239177 pixels
```

```
-----
Number of points in input object: 267498
```

```
Fraction of object selected: 100.00% (= 267498 points) ; Access speed: 1.75GB/sec
```

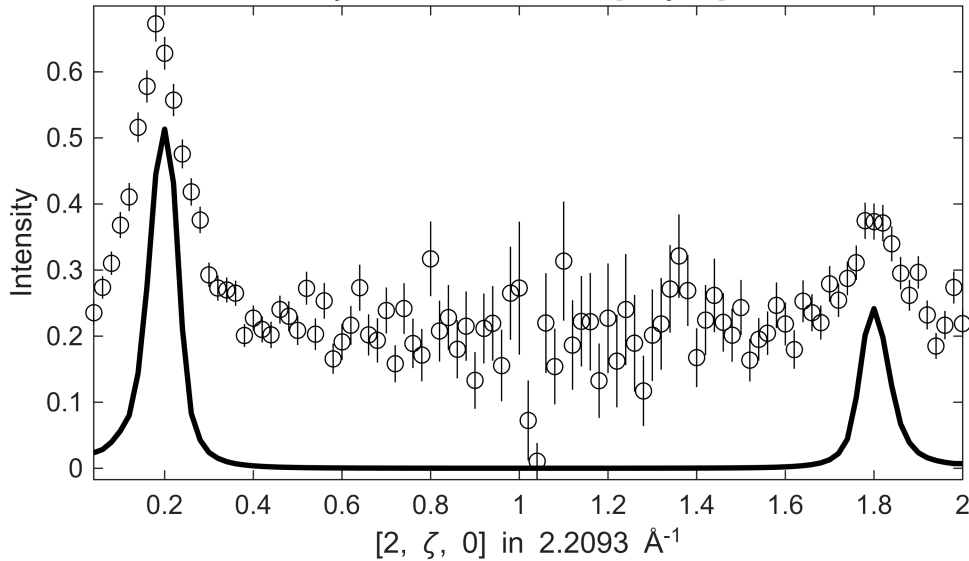
```
Fraction of object retained: 89.41% (= 239177 points)
```

```
**** Total time in cut(sqw) : 0.0sec Including:
```

Data access time fraction: 24.9%; Transf. time frac.: 55.5%; Resulting pix preparation: 19.6%

Fe_ei200_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $30 \leq E \leq 50$
 $\zeta = -0.01:0.02:2.01$ in $[2, \zeta, 0]$



**** En = 50±10

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 424838 pixels -- processing data... -----> included 390684 pixels

Number of points in input object: 424838

Fraction of object selected: 100.00% (= 424838 points) ; Access speed: 1.52GB/sec

Fraction of object retained: 91.96% (= 390684 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 51.4%; Transf. time frac.: 47.0%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 424838 pixels -- processing data... -----> retained 390684 pixels

Number of points in input object: 424838

Fraction of object selected: 100.00% (= 424838 points) ; Access speed: 2.02GB/sec

Fraction of object retained: 91.96% (= 390684 points)

**** Total time in cut(sqw) : 0.1sec Including:

Data access time fraction: 24.6%; Transf. time frac.: 51.9%; Resulting pix preparation: 23.4%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 424838 pixels -- processing data... -----> retained 390684 pixels

Number of points in input object: 424838

Fraction of object selected: 100.00% (= 424838 points) ; Access speed: 0.89GB/sec

Fraction of object retained: 91.96% (= 390684 points)

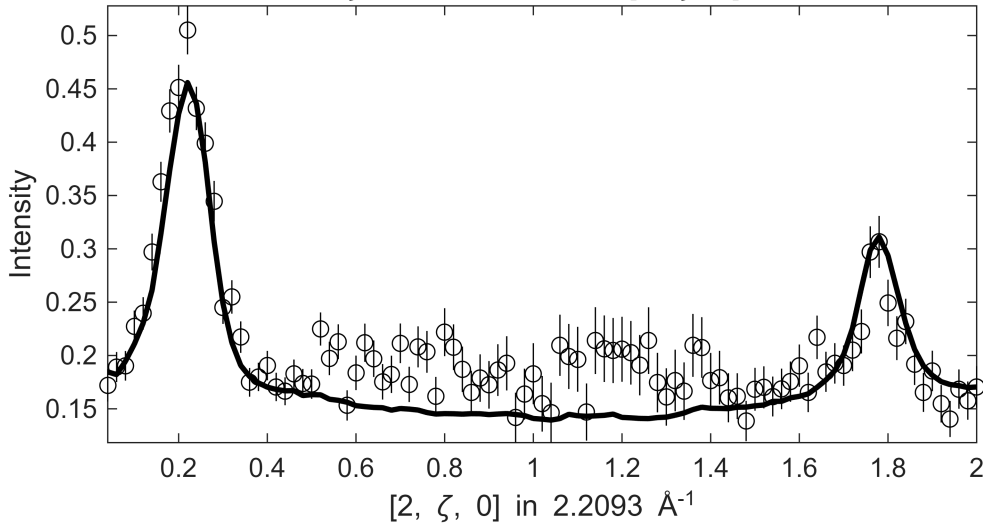
**** Total time in cut(sqw) : 0.1sec Including:

Data access time fraction: 40.9%; Transf. time frac.: 43.2%; Resulting pix preparation: 16.0%

Fe_ei200_align_sym3.sqw

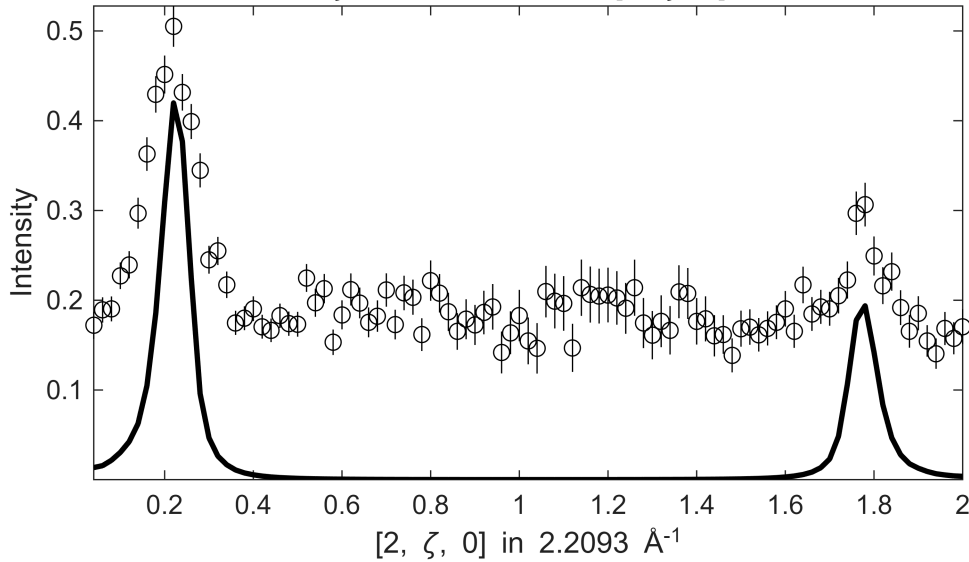
S=0.59±0.037; J0= 24±0.38; gamma=9.37308±1.46433

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 40 ≤ E ≤ 60
ζ=-0.01:0.02:2.01 in [2, ζ, 0]



Fe_ei200_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 40 ≤ E ≤ 60
ζ=-0.01:0.02:2.01 in [2, ζ, 0]



**** En = 60±10

Warning: Dataset:3 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.27003

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 524639 pixels -- processing data... -----> included 476345 pixels

Number of points in input object: 524639

Fraction of object selected: 100.00% (= 524639 points) ; Access speed: 1.78GB/sec

Fraction of object retained: 90.79% (= 476345 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 52.6%; Transf. time frac.: 45.7%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 523881 pixels -- processing data... -----> retained 475833 pixels

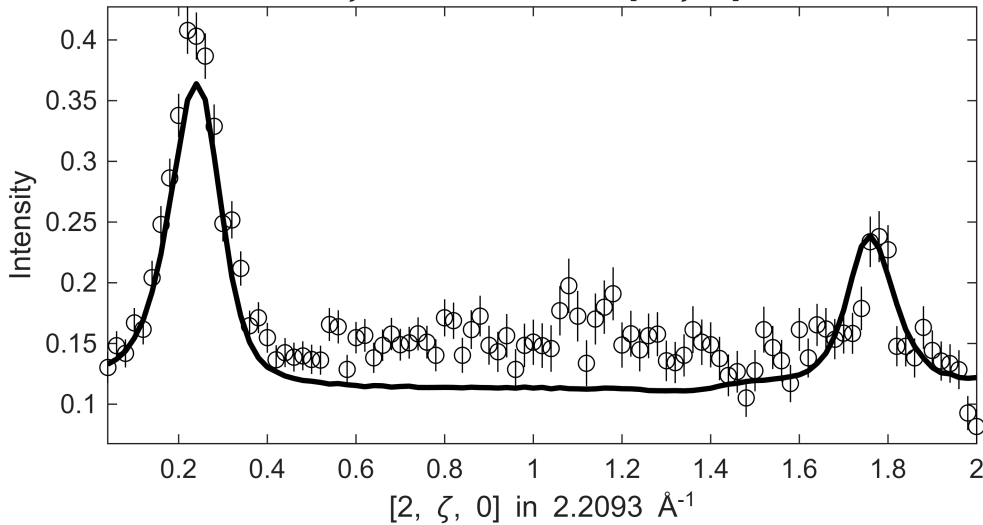
Number of points in input object: 523881

Fraction of object selected: 100.00% (= 523881 points) ; Access speed: 1.71GB/sec

Fraction of object retained: 90.83% (= 475833 points)

**** Total time in cut(sqw) : 0.1sec Including:
 Data access time fraction: 28.5%; Transf. time frac.: 47.6%; Resulting pix preparation: 23.9%

Fe_ei200_align_sym3.sqw
 $S=0.73\pm0.05$; $J_0=26\pm0.51$; $\gamma=18.457\pm2.37386$
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $50 \leq E \leq 70$
 $\zeta=-0.01:0.02:2.01$ in $[2, \zeta, 0]$

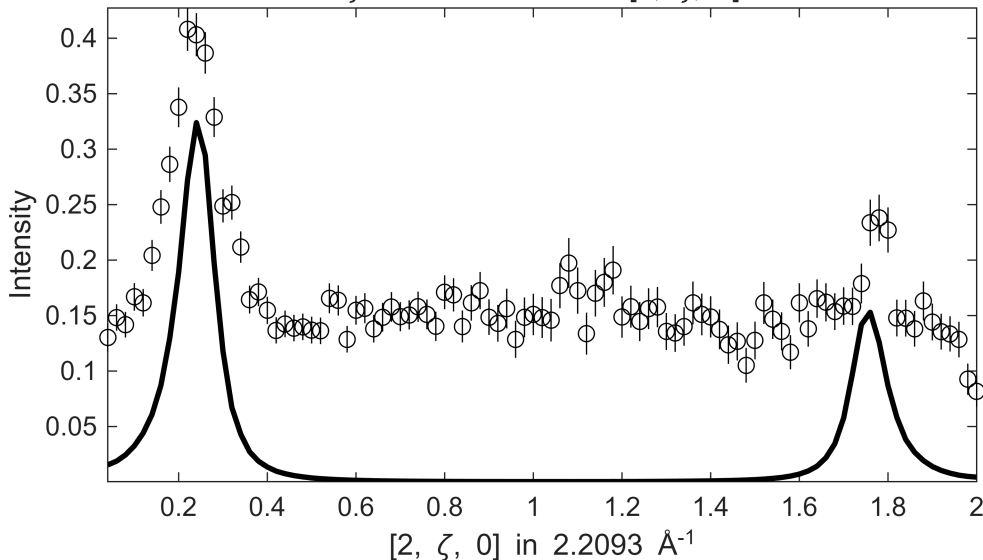


*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
 *** Step 1 of 1; Read data for 524639 pixels -- processing data... -----> retained 476345 pixels

Number of points in input object: 524639
 Fraction of object selected: 100.00% (= 524639 points); Access speed: 2.02GB/sec
 Fraction of object retained: 90.79% (= 476345 points)

**** Total time in cut(sqw) : 0.1sec Including:
 Data access time fraction: 25.9%; Transf. time frac.: 53.5%; Resulting pix preparation: 20.5%

Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $50 \leq E \leq 70$
 $\zeta=-0.01:0.02:2.01$ in $[2, \zeta, 0]$



**** $E_n = 70\pm10$

Warning: Dataset:7 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.63006

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 569894 pixels -- processing data... -----> included 517638 pixels

Number of points in input object: 569894

Fraction of object selected: 100.00% (= 569894 points) ; Access speed: 1.81GB/sec
Fraction of object retained: 90.83% (= 517638 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 50.9%; Transf. time frac.: 47.4%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 567815 pixels -- processing data... -----> retained 515559 pixels

Number of points in input object: 567815

Fraction of object selected: 100.00% (= 567815 points) ; Access speed: 1.69GB/sec
Fraction of object retained: 90.80% (= 515559 points)

**** Total time in cut(sqw) : 0.1sec Including:

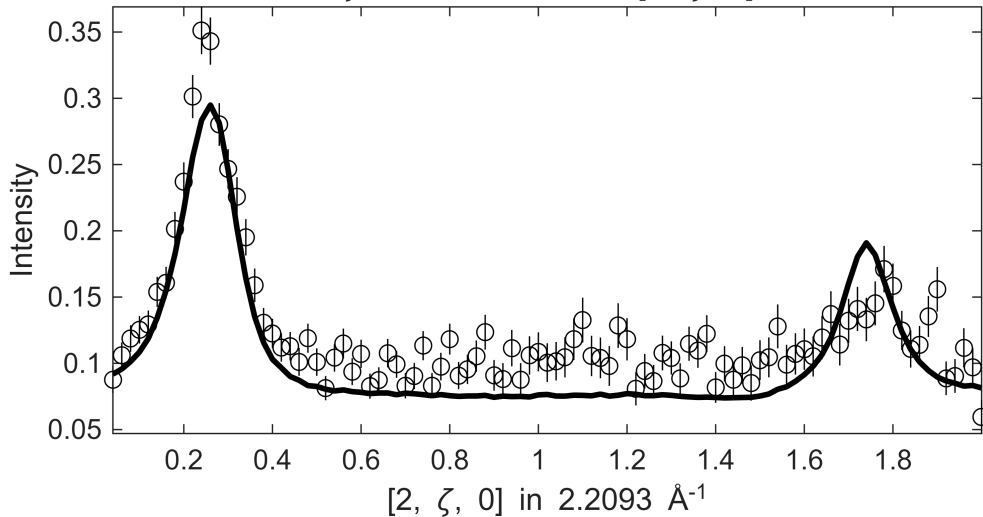
Data access time fraction: 26.0%; Transf. time frac.: 53.9%; Resulting pix preparation: 20.1%

Fe_ei200_align_sym3.sqw

S= 0.8±0.022; J0= 27± 0.4; gamma=26.0733±0.644627

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 60 ≤ E ≤ 80

ζ=-0.01:0.02:2.01 in [2, ζ, 0]



*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 569894 pixels -- processing data... -----> retained 517638 pixels

Number of points in input object: 569894

Fraction of object selected: 100.00% (= 569894 points) ; Access speed: 1.95GB/sec
Fraction of object retained: 90.83% (= 517638 points)

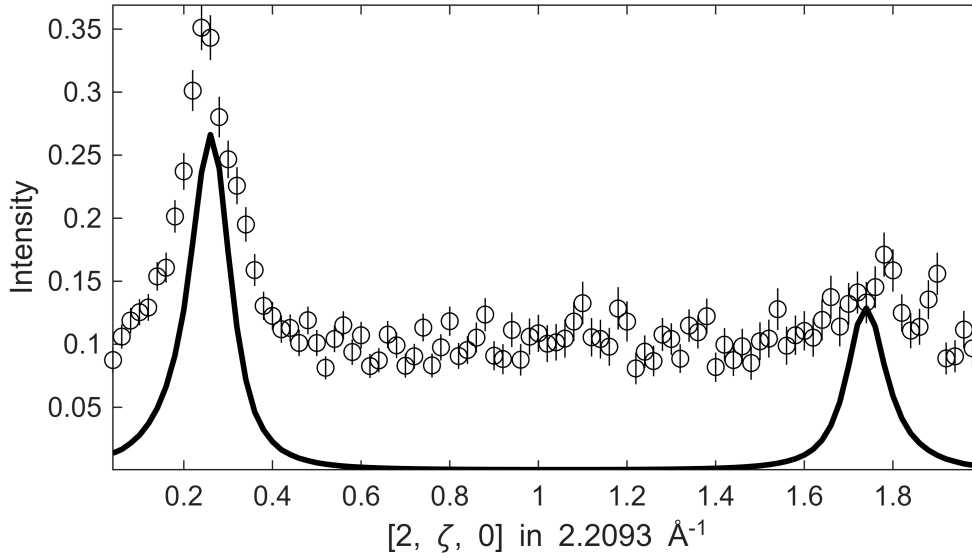
**** Total time in cut(sqw) : 0.1sec Including:

Data access time fraction: 26.7%; Transf. time frac.: 52.0%; Resulting pix preparation: 21.3%

Fe_ei200_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $60 \leq E \leq 80$

$\zeta = -0.01:0.02:2.01$ in $[2, \zeta, 0]$



**** En = 80±10

Warning: Dataset:19 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.7102

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 632375 pixels -- processing data... -----> included 573852 pixels

Number of points in input object: 632375

Fraction of object selected: 100.00% (= 632375 points) ; Access speed: 1.67GB/sec

Fraction of object retained: 90.75% (= 573852 points)

**** Total time in cut(sqw) : 0.1sec Including:

Data access time fraction: 50.9%; Transf. time frac.: 47.1%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 626326 pixels -- processing data... -----> retained 568383 pixels

Number of points in input object: 626326

Fraction of object selected: 100.00% (= 626326 points) ; Access speed: 2.04GB/sec

Fraction of object retained: 90.75% (= 568383 points)

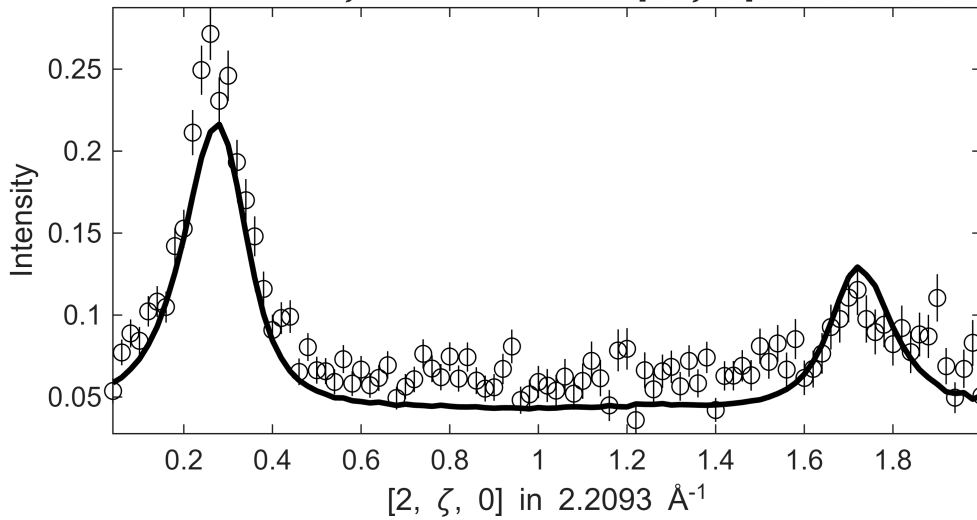
**** Total time in cut(sqw) : 0.1sec Including:

Data access time fraction: 20.2%; Transf. time frac.: 55.9%; Resulting pix preparation: 23.9%

Fe_ei200_align_sym3.sqw

S= 0.9±0.067; J0= 30±0.84; gamma=42.1798±5.05833

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 70 ≤ E ≤ 90
ζ=-0.01:0.02:2.01 in [2, ζ, 0]

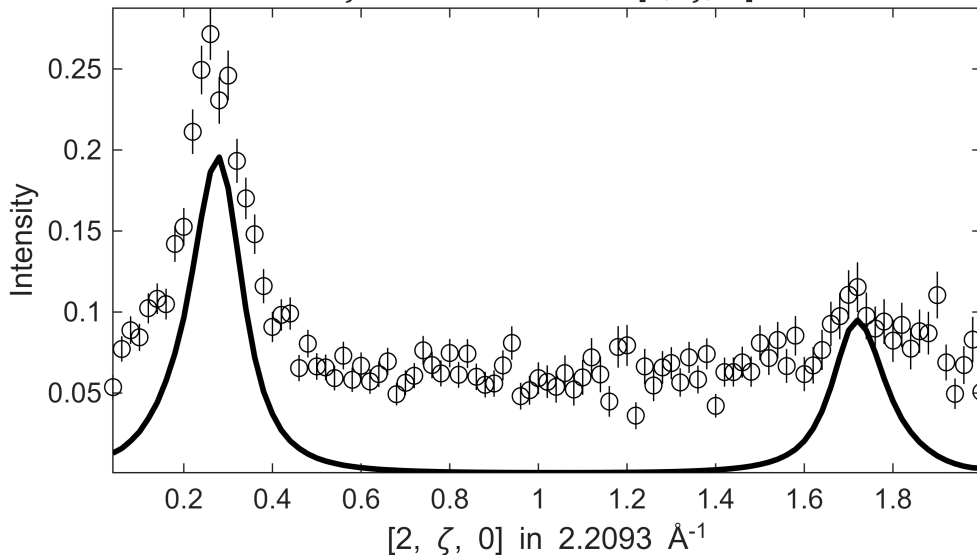


```
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 632375 pixels -- processing data... -----> retained 573852 pixels
-----
Number of points in input object: 632375
  Fraction of object selected: 100.00% (= 632375 points) ; Access speed: 1.76GB/sec
  Fraction of object retained: 90.75% (= 573852 points)

**** Total time in cut(sqw) : 0.1sec Including:
    Data access time fraction: 29.9%; Transf. time frac.: 49.7%; Resulting pix preparation: 20.5%
-----
```

Fe_ei200_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 70 ≤ E ≤ 90
ζ=-0.01:0.02:2.01 in [2, ζ, 0]



```
*****
**** En = 90±10
*****
Warning: Dataset:36 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 3.2403
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 712163 pixels -- processing data... -----> included 647739 pixels
-----
Number of points in input object: 712163
  Fraction of object selected: 100.00% (= 712163 points) ; Access speed: 2.03GB/sec
  Fraction of object retained: 90.95% (= 647739 points)
```

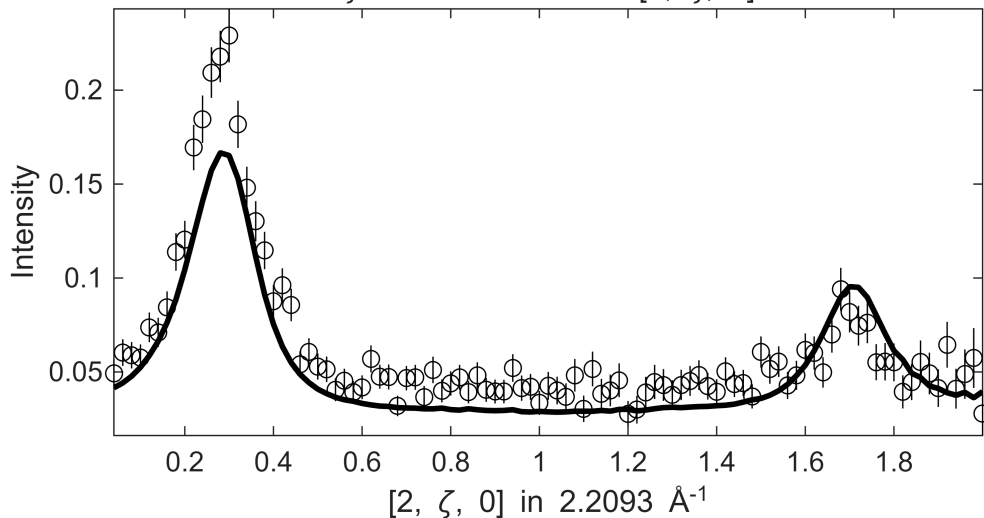
```

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 47.2%; Transf. time frac.: 51.3%; Resulting pix preparation:  0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 696901 pixels -- processing data... -----> retained 633307 pixels
-----
Number of points in input object: 696901
      Fraction of object selected: 100.00% (= 696901 points) ; Access speed:      1.40GB/sec
      Fraction of object retained: 90.87% (= 633307 points)

**** Total time in cut(sqw)   :      0.1sec Including:
      Data access time fraction: 34.9%; Transf. time frac.: 45.6%; Resulting pix preparation: 19.5%
-----

```

Fe_ei200_align_sym3.sqw
 $S=0.89\pm0.074$; $J_0=32\pm1.1$; $\gamma=55.1597\pm6.97689$
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $80 \leq E \leq 10$
 $\zeta=-0.01:0.02:2.01$ in $[2, \zeta, 0]$



```

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 712163 pixels -- processing data... -----> retained 647739 pixels
-----
Number of points in input object: 712163
      Fraction of object selected: 100.00% (= 712163 points) ; Access speed:      2.04GB/sec
      Fraction of object retained: 90.95% (= 647739 points)

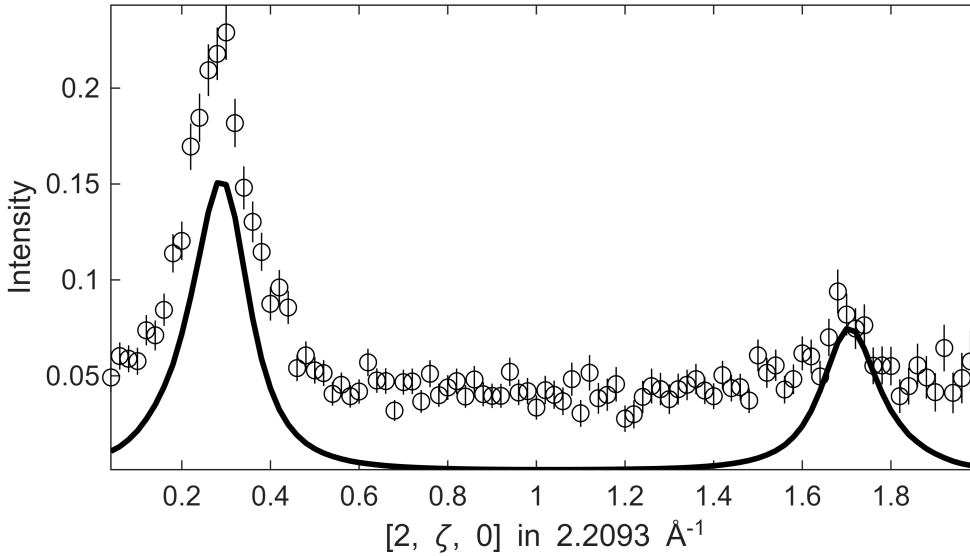
**** Total time in cut(sqw)   :      0.1sec Including:
      Data access time fraction: 28.2%; Transf. time frac.: 50.9%; Resulting pix preparation: 20.9%
-----

```

Fe_ei200_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $80 \leq E \leq 10$

$\zeta = -0.01:0.02:2.01$ in $[2, \zeta, 0]$



**** En = 100±10

Warning: Dataset:75 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 6.7507

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 781729 pixels -- processing data... -----> included 710069 pixels

Number of points in input object: 781729

Fraction of object selected: 100.00% (= 781729 points) ; Access speed: 1.98GB/sec

Fraction of object retained: 90.83% (= 710069 points)

**** Total time in cut(sqw) : 0.1sec Including:

Data access time fraction: 45.5%; Transf. time frac.: 53.3%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 750621 pixels -- processing data... -----> retained 681449 pixels

Number of points in input object: 750621

Fraction of object selected: 100.00% (= 750621 points) ; Access speed: 1.46GB/sec

Fraction of object retained: 90.78% (= 681449 points)

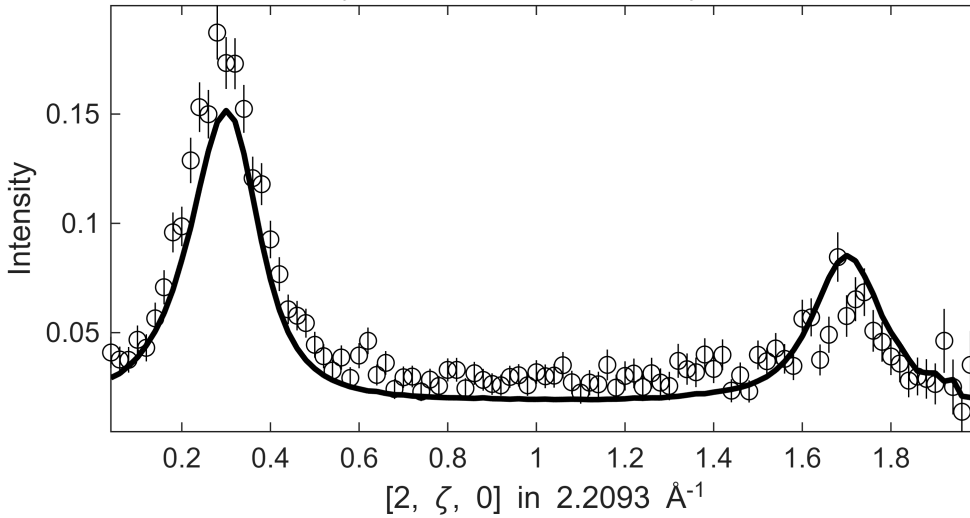
**** Total time in cut(sqw) : 0.1sec Including:

Data access time fraction: 32.9%; Transf. time frac.: 47.7%; Resulting pix preparation: 19.4%

Fe_ei200_align_sym3.sqw

S=0.92±0.064; J0= 32±0.91; gamma=59.8613±6.29502

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 90 ≤ E ≤ 110
ζ=-0.01:0.02:2.01 in [2, ζ, 0]

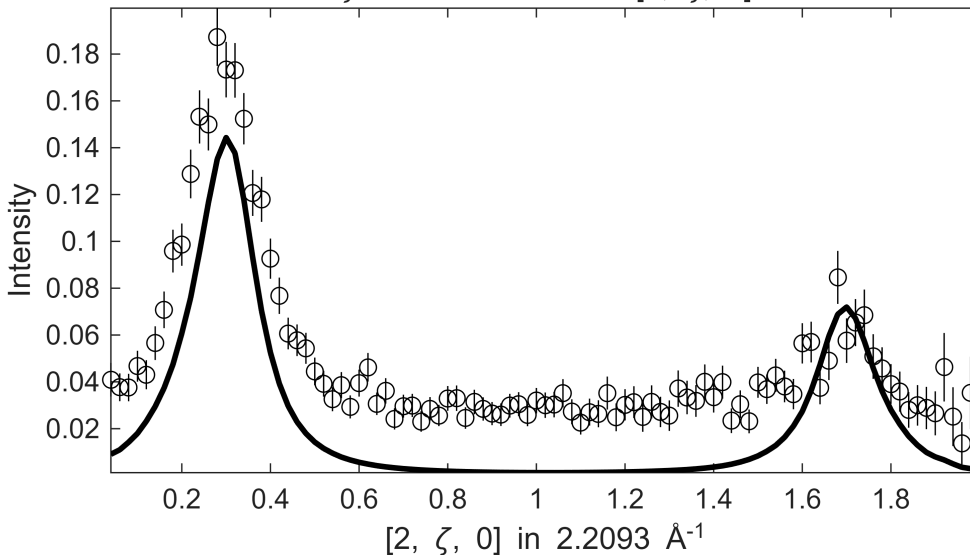


```
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 781729 pixels -- processing data... -----> retained 710069 pixels
-----
Number of points in input object: 781729
  Fraction of object selected: 100.00% (= 781729 points) ; Access speed: 1.71GB/sec
  Fraction of object retained: 90.83% (= 710069 points)

**** Total time in cut(sqw) : 0.1sec Including:
    Data access time fraction: 30.4%; Transf. time frac.: 49.6%; Resulting pix preparation: 20.0%
-----
```

Fe_ei200_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 90 ≤ E ≤ 110
ζ=-0.01:0.02:2.01 in [2, ζ, 0]



```
*****
**** En = 110±10
*****
Warning: Dataset:108 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 9.721
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 954457 pixels -- processing data... -----> included 854010 pixels
-----
Number of points in input object: 954457
  Fraction of object selected: 100.00% (= 954457 points) ; Access speed: 2.03GB/sec
  Fraction of object retained: 89.48% (= 854010 points)
```

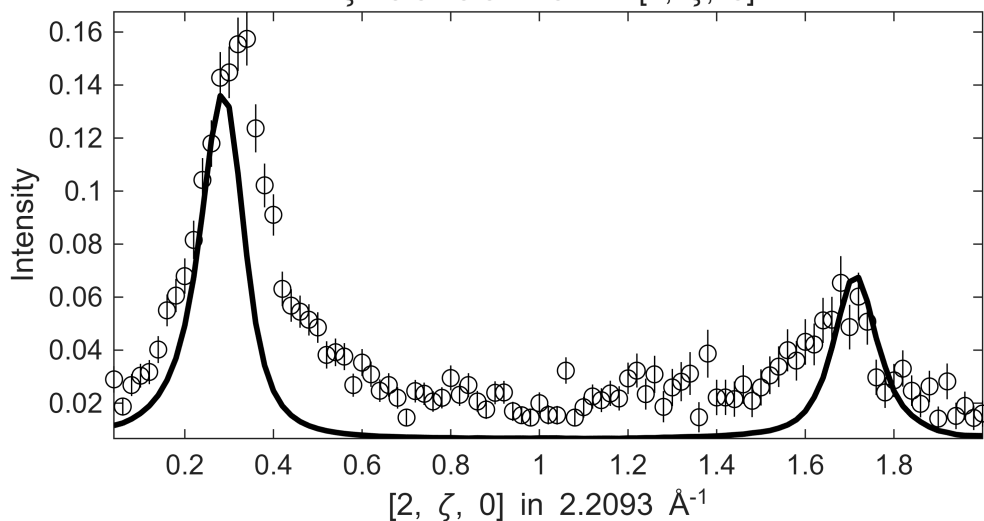
```

**** Total time in cut(sqw)   :      0.1sec Including:
      Data access time fraction:  49.5%; Transf. time frac.:  49.6%; Resulting pix preparation:  0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for  908088 pixels -- processing data... -----> retained  813644 pixels
-----
Number of points in input object: 908088
      Fraction of object selected:  100.00% (=  908088 points) ; Access speed:      2.05GB/sec
      Fraction of object retained:   89.60% (=  813644 points)

**** Total time in cut(sqw)   :      0.1sec Including:
      Data access time fraction:  27.3%; Transf. time frac.:  52.7%; Resulting pix preparation:  20.0%
-----

```

Fe_ei200_align_sym3.sqw
 $S=0.54\pm0.031$; $J_0= 36\pm0.62$; $\gamma=31.0731\pm3.08751$
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $100 \leq E \leq 12$
 $\zeta=-0.01:0.02:2.01$ in $[2, \zeta, 0]$



```

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for  954457 pixels -- processing data... -----> retained  854010 pixels
-----
Number of points in input object: 954457
      Fraction of object selected:  100.00% (=  954457 points) ; Access speed:      1.99GB/sec
      Fraction of object retained:   89.48% (=  854010 points)

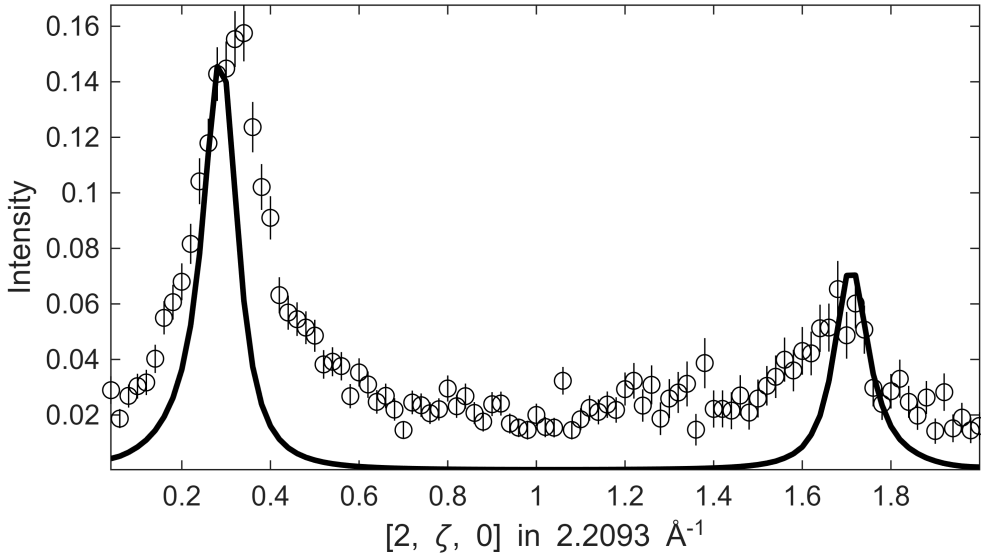
**** Total time in cut(sqw)   :      0.1sec Including:
      Data access time fraction:  30.2%; Transf. time frac.:  48.5%; Resulting pix preparation:  21.3%
-----

```

Fe_ei200_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $100 \leq E \leq 12$

$\zeta = -0.01:0.02:2.01$ in $[2, \zeta, 0]$



**** En = 120±10

Warning: Dataset:89 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 8.0108

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 1569314 pixels -- processing data... -----> included 1403471 pixels

Number of points in input object: 1569314

Fraction of object selected: 100.00% (= 1569314 points) ; Access speed: 1.82GB/sec

Fraction of object retained: 89.43% (= 1403471 points)

**** Total time in cut(sqw) : 0.1sec Including:

Data access time fraction: 57.9%; Transf. time frac.: 41.3%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 1518162 pixels -- processing data... -----> retained 1355430 pixels

Number of points in input object: 1518162

Fraction of object selected: 100.00% (= 1518162 points) ; Access speed: 2.03GB/sec

Fraction of object retained: 89.28% (= 1355430 points)

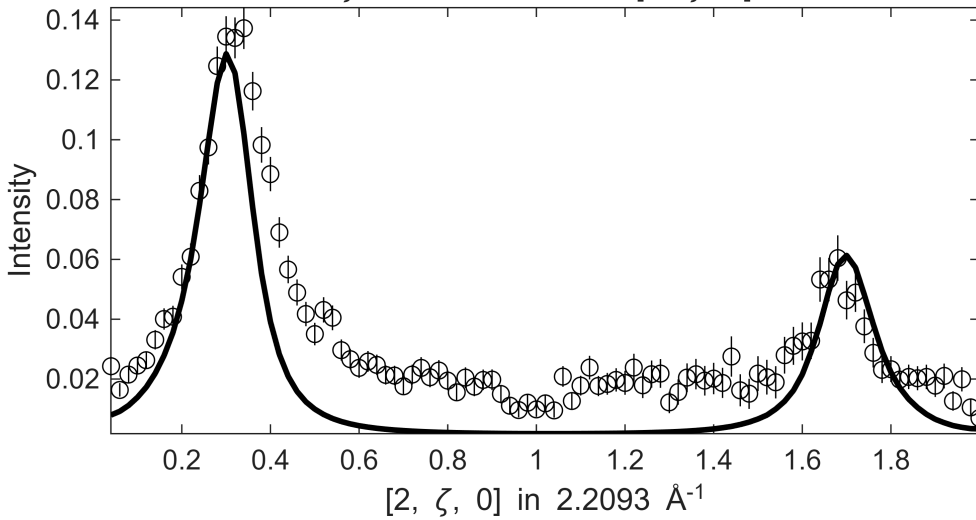
**** Total time in cut(sqw) : 0.2sec Including:

Data access time fraction: 32.4%; Transf. time frac.: 45.7%; Resulting pix preparation: 21.9%

Fe_ei200_align_sym3.sqw

S=0.74±0.031; J0= 37±0.59; gamma=49.9248±3.587

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 110 ≤ E ≤ 13
ζ=-0.01:0.02:2.01 in [2, ζ, 0]

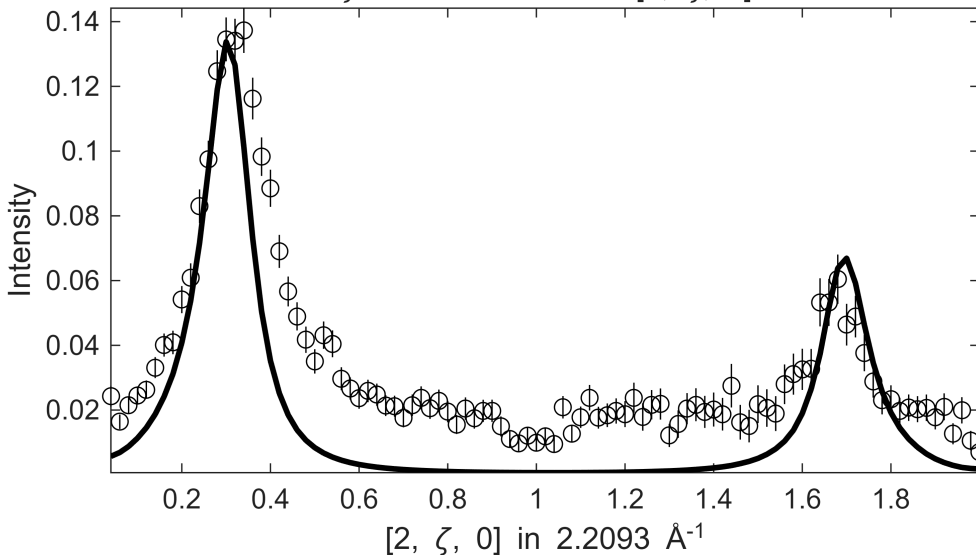


```
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 1569314 pixels -- processing data... -----> retained 1403471 pixels
-----
Number of points in input object: 1569314
  Fraction of object selected: 100.00% (= 1569314 points) ; Access speed: 1.52GB/sec
  Fraction of object retained: 89.43% (= 1403471 points)

**** Total time in cut(sqw) : 0.2sec Including:
    Data access time fraction: 36.9%; Transf. time frac.: 44.3%; Resulting pix preparation: 18.8%
-----
```

Fe_ei200_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 110 ≤ E ≤ 13
ζ=-0.01:0.02:2.01 in [2, ζ, 0]



```
*****
**** En = 130±10
*****
Warning: Dataset:59 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 5.3105
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 2880160 pixels -- processing data... -----> included 2621036 pixels
-----
Number of points in input object: 2880160
  Fraction of object selected: 100.00% (= 2880160 points) ; Access speed: 2.07GB/sec
  Fraction of object retained: 91.00% (= 2621036 points)
```

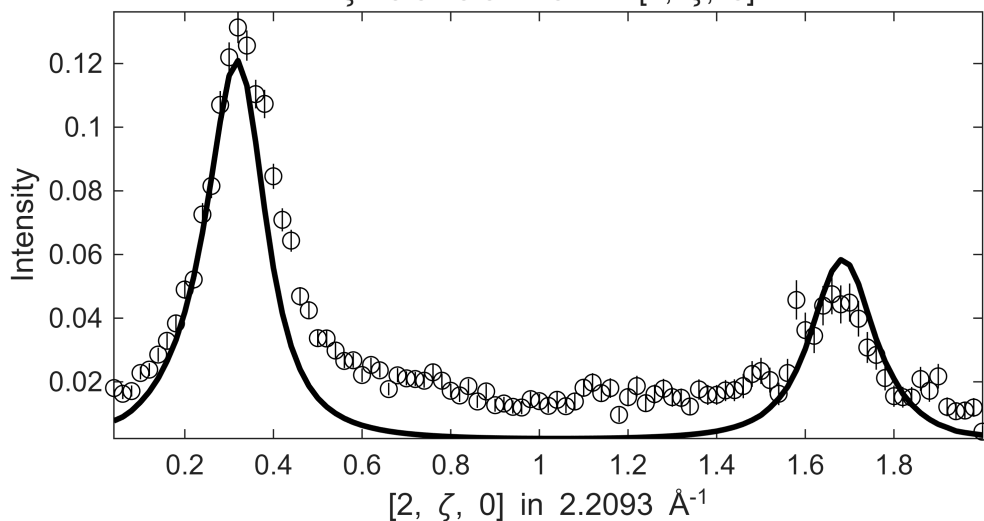
```

**** Total time in cut(sqw)   :      0.1sec Including:
      Data access time fraction:  62.5%; Transf. time frac.:  37.0%; Resulting pix preparation:  0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for  2823941 pixels -- processing data... -----> retained  2572404 pixels
-----
Number of points in input object: 2823941
      Fraction of object selected:  100.00% (=  2823941 points) ; Access speed:      1.59GB/sec
      Fraction of object retained:   91.09% (=  2572404 points)

**** Total time in cut(sqw)   :      0.3sec Including:
      Data access time fraction:  36.0%; Transf. time frac.:  44.2%; Resulting pix preparation:  19.8%
-----

```

Fe_ei200_align_sym3.sqw
 $S=0.86\pm0.034$; $J_0= 38\pm0.66$; $\gamma=64.1485\pm4.31739$
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $120 \leq E \leq 140$ eV
 $\zeta=-0.01:0.02:2.01$ in $[2, \zeta, 0]$



```

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for  2880160 pixels -- processing data... -----> retained  2621036 pixels
-----
Number of points in input object: 2880160
      Fraction of object selected:  100.00% (=  2880160 points) ; Access speed:      1.69GB/sec
      Fraction of object retained:   91.00% (=  2621036 points)

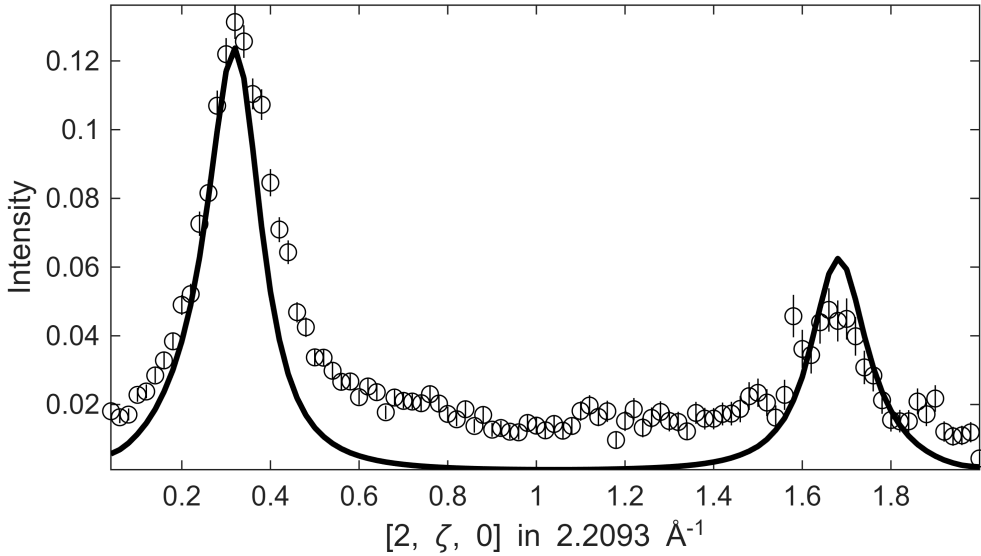
**** Total time in cut(sqw)   :      0.3sec Including:
      Data access time fraction:  36.0%; Transf. time frac.:  44.0%; Resulting pix preparation:  20.1%
-----

```


Fe_ei200_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $120 \leq E \leq 140$

$\zeta = -0.01:0.02:2.01$ in $[2, \zeta, 0]$



**** En = 140±10

Warning: Dataset:45 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 4.0504

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 3595796 pixels -- processing data... -----> included 3345767 pixels

Number of points in input object: 3595796

Fraction of object selected: 100.00% (= 3595796 points) ; Access speed: 1.72GB/sec

Fraction of object retained: 93.05% (= 3345767 points)

**** Total time in cut(sqw) : 0.2sec Including:

Data access time fraction: 65.7%; Transf. time frac.: 33.2%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 3541377 pixels -- processing data... -----> retained 3295882 pixels

Number of points in input object: 3541377

Fraction of object selected: 100.00% (= 3541377 points) ; Access speed: 1.81GB/sec

Fraction of object retained: 93.07% (= 3295882 points)

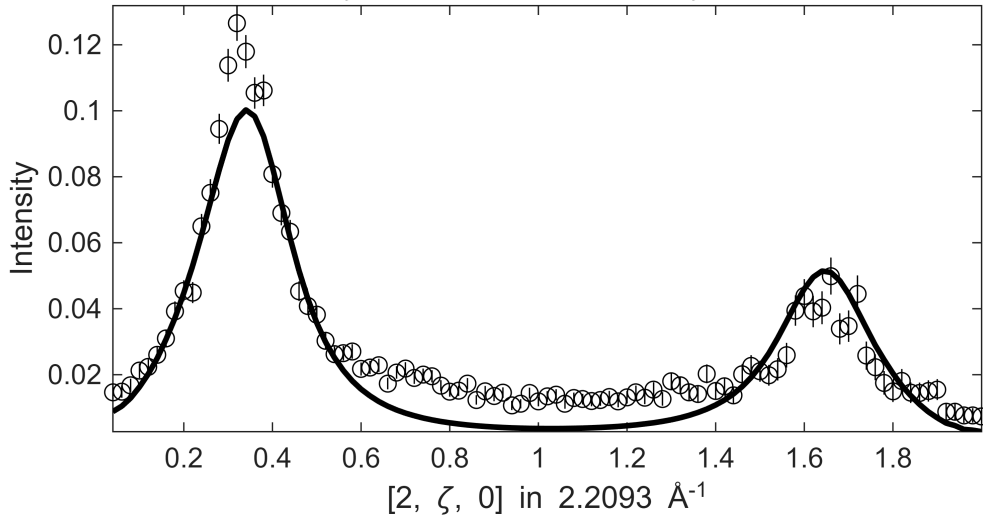
**** Total time in cut(sqw) : 0.4sec Including:

Data access time fraction: 34.5%; Transf. time frac.: 42.8%; Resulting pix preparation: 22.6%

Fe_ei200_align_sym3.sqw

S= 1.6±0.13; J0= 43± 2.2; gamma=164.081±21.6541

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 130 ≤ E ≤ 15
ζ=-0.01:0.02:2.01 in [2, ζ, 0]



*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 3595796 pixels -- processing data... -----> retained 3345767 pixels

Number of points in input object: 3595796

Fraction of object selected: 100.00% (= 3595796 points) ; Access speed: 1.93GB/sec

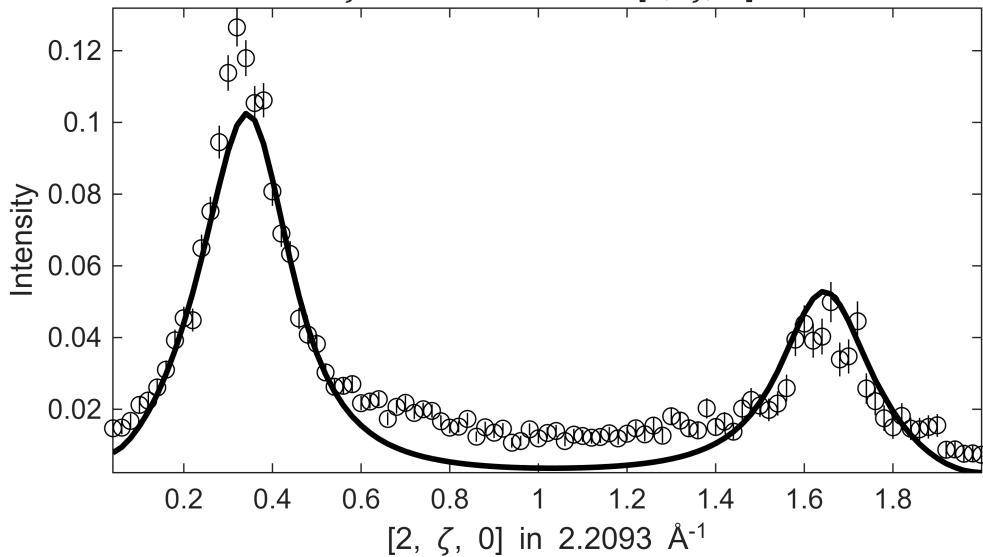
Fraction of object retained: 93.05% (= 3345767 points)

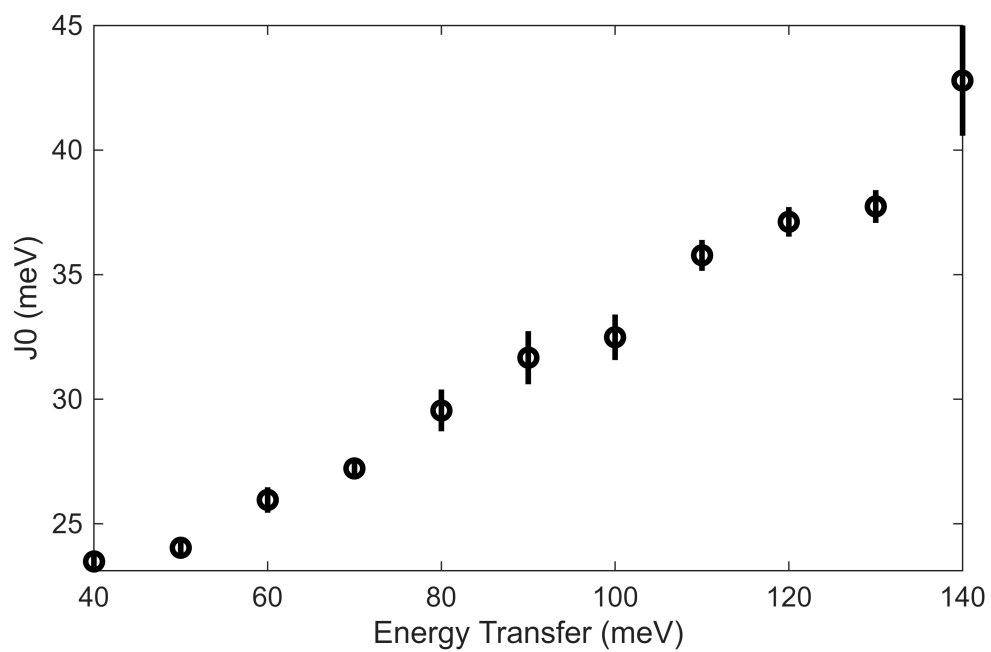
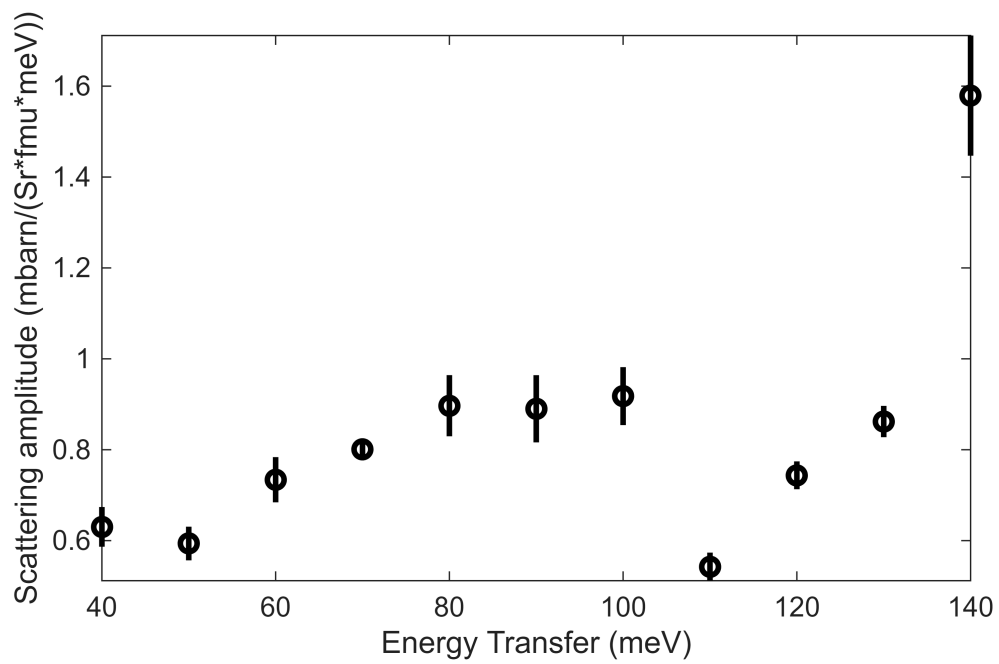
**** Total time in cut(sqw) : 0.4sec Including:

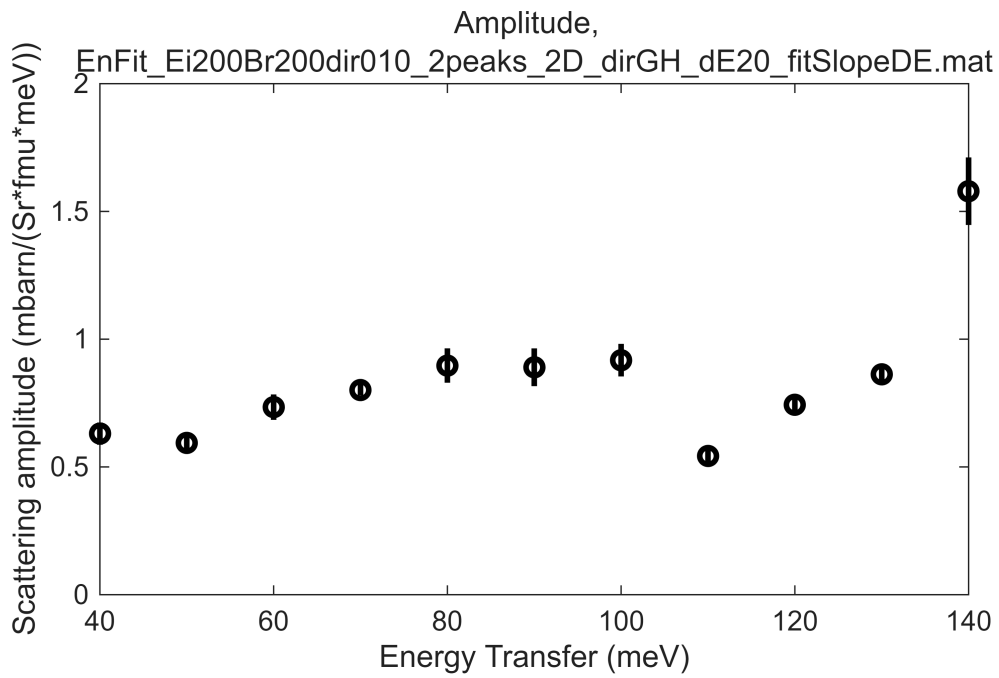
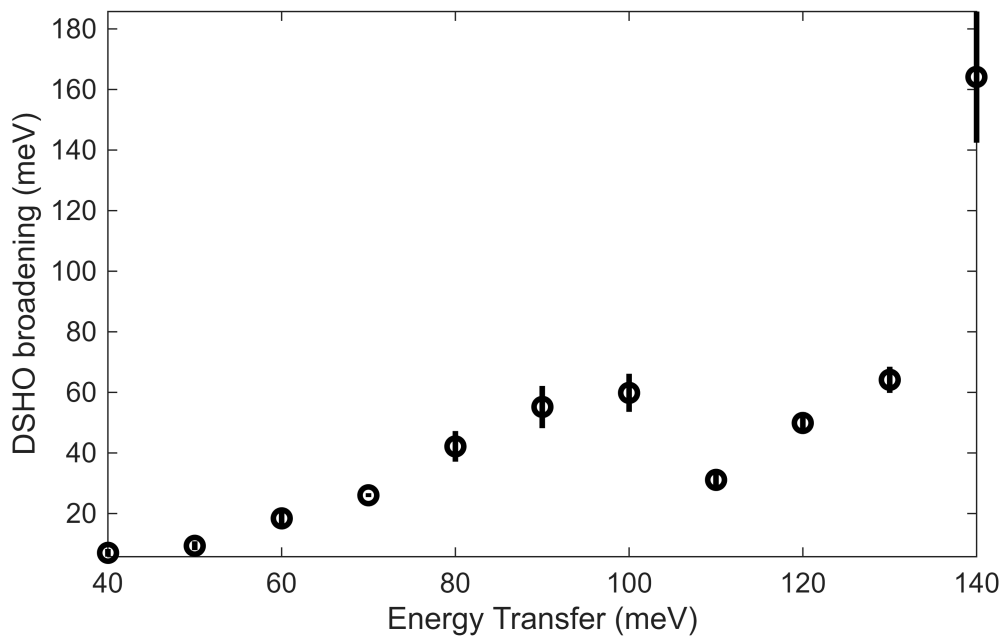
Data access time fraction: 32.1%; Transf. time frac.: 45.8%; Resulting pix preparation: 22.1%

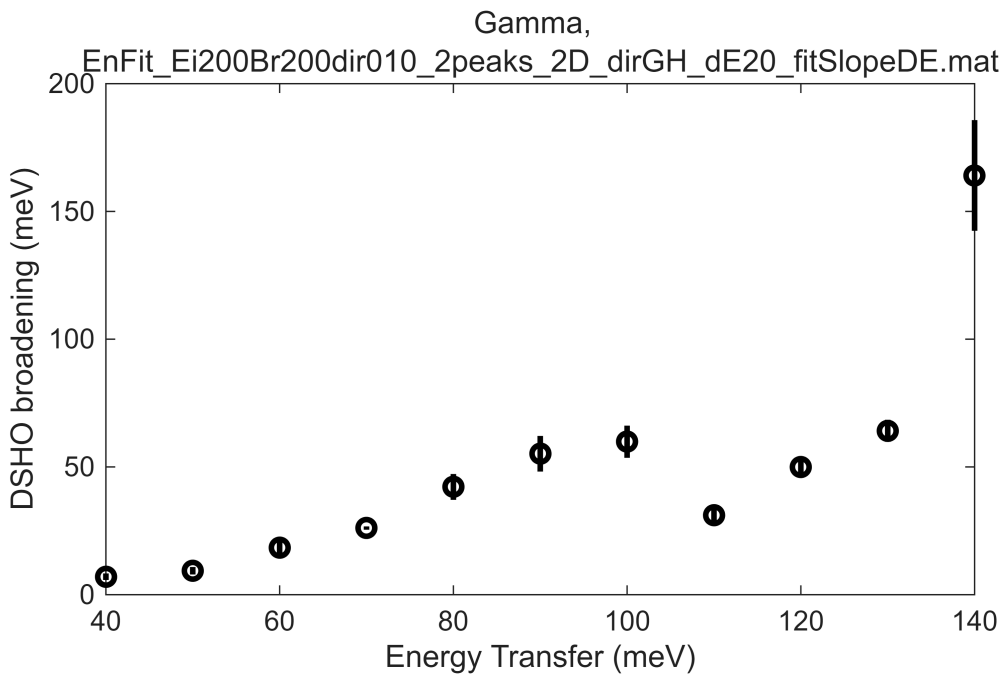
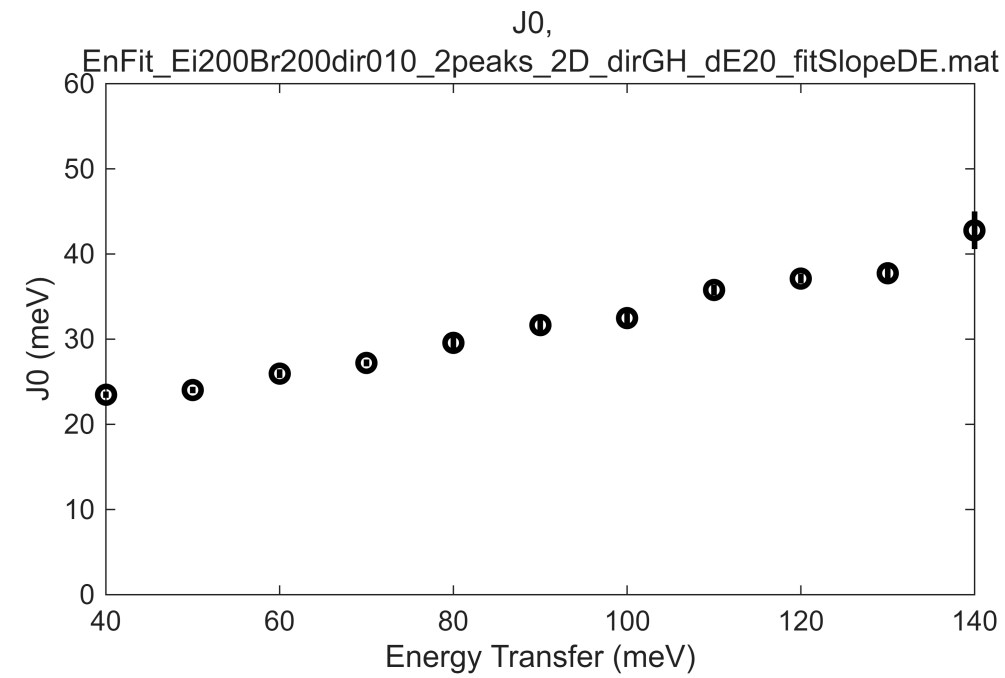
Fe_ei200_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 130 ≤ E ≤ 15
ζ=-0.01:0.02:2.01 in [2, ζ, 0]



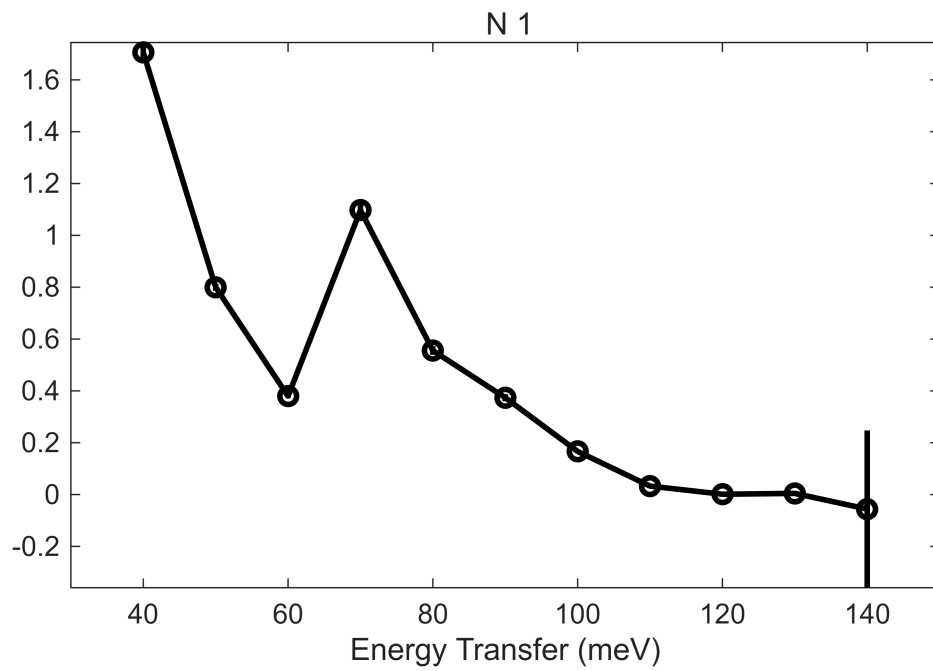




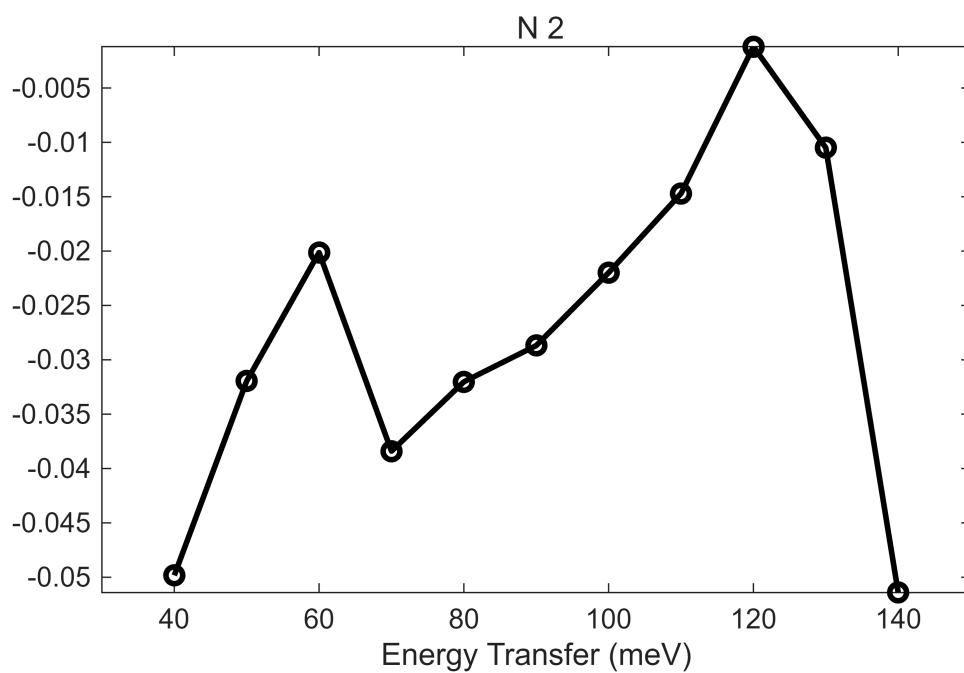


beep
beep
beep

```
bg_data = extract_bg_par(fit_res_200fix_bg.all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_f
figure;
```



```
pd(bg_data(2));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_f
figure
```



```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_
figure;
```

Index exceeds the number of array elements. Index must not exceed 2.

```
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_
figure
```

```
fit_resEi200 = fit_res_200fix_bg;
```

```
save(res_name, "fit_resEi200")
```